

VARUN KUMAR H C

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Objective

Motivated B.E graduate in Artificial Intelligence and Machine Learning. Seeking an entry-level role where I can apply my technical knowledge, learn industry practices, and contribute to building scalable and efficient technology solutions.

Internships & Training

Data Scientist Intern – AIML LABS PVT LTD (April 2025 – Present)

- Completed hands-on training in Data Science and Machine Learning with a focus on real-world problem solving and model development.
- Gained practical experience in data preprocessing, EDA, and feature engineering techniques.
- Built and evaluated multiple machine learning models using classification algorithms and metrics
- Worked on advanced concepts including Generative AI, Retrieval-Augmented Generation (RAG), and LLM-based applications using LangChain and vector databases

Skills

Programming: Python (Pandas, NumPy, Matplotlib, Seaborn) ,SQL

Data Visualization: Power BI, Tableau

Machine Learning: Supervised Learning , Unsupervised Learning

Statistics & Analytics: EDA ,

Generative AI & NLP:LangChain, LLMs (GPT-3.5, Mistral),RAG, Prompt Engineering, Sentence Transformers

Deployment & Tools: Streamlit, Render, Docker (Basie), Azure, Kubernetes (Basic)

Education

BE/BTECH (Artificial Intelligence & Machine Learning)	2021 -2025	SEA College of Engineering, Bengaluru	CGPA – 7.01
PUC (Class XII)	2021	Shree Vijetha PU College, Chintamani	71.5%
SSLC (Class X)	2019	Nalanda Vidhya Mandira ,Chintamani	68.18%

Generative AI Chatbot for Customer Support Automation

Gen AI Project

Title: Generative AI Chatbot for Customer Support Automation

Description: Developed a Generative AI-powered customer support chatbot using a Retrieval Augmented Generation (RAG) architecture to deliver context-aware, accurate responses to customer queries by leveraging internal knowledge base documents and FAQs.

Skills Used: Generative AI, RAG, LangChain, LLMs (GPT-3.5/Mistral), Prompt Engineering, Sentence Transformers, Vector Databases (ChromaDB), Python, Flask, NLP

Roles and Responsibilities:

- Designed and implemented a Generative AI chatbot using RAG architecture to provide automated, context-aware responses to customer inquiries based on company knowledge base.
- Utilized LangChain to orchestrate the workflow, integrating document loaders, text splitters, embedding models (e.g., SentenceTransformers), and a Vector Database (ChromaDB) for efficient information retrieval.
- Fine-tuned prompts for an LLM (e.g., GPT-3.5/Mistral) to generate accurate and relevant answers based on retrieved context, improving response quality and consistency.
- Developed a simple Flask interface for interaction and testing of the chatbot functionality.

Customer Churn Prediction in Retail

ML Project

Title: Customer Churn Prediction in Retail

Description: Designed a customer churn prediction model for a retail loyalty program using historical purchase data, engagement metrics, and promotional responses. The model identified high-risk churn customers, enabling targeted retention strategies and reducing churn by 15%.

Skills Used: Machine Learning, XGBoost, Decision Trees, Random Forest, Feature Engineering, Data Visualization, Azure ML

Roles and Responsibilities:

- Collected and processed transactional and customer engagement data for feature extraction and modeling.
- Designed and implemented machine learning models using XGBoost, Decision Trees, and Random Forest to predict churn.
- Deployed the model using Azure ML to ensure scalability and real-time predictions.
- Developed a dashboard for customer retention teams to visualize churn risk and customer segmentation.
- Conducted statistical analysis and A/B testing to measure the effectiveness of retention campaigns.
- Integrated the model into the CRM system to automate personalized retention strategies based on churn scores.

Description:

Built a machine learning model to predict heart disease risk using patient clinical data, with end-to-end preprocessing, model comparison, and deployment as a web app.

Skills:

Python, Scikit-learn, EDA, Feature Engineering, Streamlit, Model Evaluation, Render

Key Contributions:

- Performed EDA and preprocessing (encoding, scaling using StandardScaler)
- Trained and compared ML models: Logistic Regression, KNN, SVM, Naive Bayes, Decision Tree
- Evaluated using Accuracy & F1 Score; selected **KNN** as best-performing model
- Developed and deployed a **Streamlit web app** for real-time predictions